

LM Evaluation

- Assume we have m sentences $s_1, s_2, \dots, s_m$
- The common way to evaluate language models is by the probability they assign to held-out data
- More exactly, we use *perplexity*:

$$\text{Perplexity} = e^{-l} \quad l = \frac{1}{M} \sum_{i=1}^m \log P(s_i) = \frac{1}{M} \sum_{i=1}^m \sum_j \log P(x_j^{(i)} | x_1^{(i)}, \dots, x_{j-1}^{(i)})$$

M is the total number of tokens (words) in the test data

m is the number of sentences in the test data

Some Perplexity Values

- Perplexity measures the effective number of possibilities for the next word (on average)
- If our model predicts the probability of a word to be $1/N$, regardless of the context (a uniform model), the perplexity will be N
- Perplexity values would change considerably between corpora, but Markov models often vary between 50 and 500. Newer models can be an order of magnitude or more lower
- **Extrinsic evaluation** (by applying it to a downstream application) is usually conducted in parallel to perplexity evaluation